05 Joins And Union

Create Table:

```
CREATE TABLE Students (StudentID INT PRIMARY KEY,StudentName VARCHAR(50));

CREATE TABLE Courses (CourseID INT PRIMARY KEY,CourseName VARCHAR(50));

CREATE TABLE Enrollments (EnrollmentID INT PRIMARY KEY,entID INT,
    CourseID INT,FOREIGN KEY (StudentID) REFERENCES Students(StudentID),
    FOREIGN KEY (CourseID) REFERENCES Courses(CourseID));

CREATE TABLE CurrentEmployees (EmpID INT PRIMARY KEY, EmpName VARCHAR(50));

CREATE TABLE PastEmployees (EmpID INT PRIMARY KEY,EmpName VARCHAR(50));
```

Tables_in_employee
courses
currentemployees
departments
employees
enrollments
location
pastemployees
students

1: INNER JOIN - Show students with their enrolled course names

```
SELECT s.StudentName, c.CourseName FROM Students s
INNER JOIN Enrollments e ON s.StudentID = e.StudentID
INNER JOIN Courses c ON e.CourseID = c.CourseID;
```

StudentName	CourseName
Alice	Mathematics
Bob	Physics
Charlie	Computer Science

2: LEFT & RIGHT JOIN -all students and their courses, including those without matches.

```
SELECT s.StudentName, c.CourseName FROM Enrollments e
RIGHT JOIN Courses c ON e.CourseID = c.CourseID
LEFT JOIN Students s ON e.StudentID = s.StudentID;
```

StudentName	CourseName
Alice	Mathematics
Bob	Physics
Charlie	Computer Science

3: UNION vs UNION ALL - Combine lists of current and past employees.

```
SELECT EmpName FROM CurrentEmployees
UNION
SELECT EmpName FROM PastEmployees;
```

EmpName
John
Mary
Steve
Paul
Emma

```
SELECT EmpName FROM CurrentEmployees
UNION ALL
SELECT EmpName FROM PastEmployees;
```

EmpName
John
Mary
Steve
Mary
Paul
Emma